

The pair singular series of a polynomial III: the off-diagonal in Cesàro form

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Abstract

[Working draft. Theorem A is the target; nothing in Sections 3–4 is proved yet.] Parts I and II reduced the conjecture $\sum_{h \leq H} (S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + A_f + o(1)$ for the pair singular series of an irreducible polynomial f to a statement about the off-diagonal sum over pairs of distinct roots of f modulo squarefree d (“Hypothesis (E)”), and showed that in Cesàro form any unbounded saving over the trivial bound suffices when the Galois group is 2-transitive. Here we prove Hypothesis (E) in Cesàro form for every irreducible quadratic, [goal], by reducing it to weighted Weyl sums of the roots of $\nu^2 \equiv D \pmod{d}$ with a phase $e(kH/d)$, in the manner of Hooley’s treatment of $\sum_{n \leq x} \tau(n^2 + a)$; the frequency k only needs to be controlled up to a power of $\log H$, because the Cesàro weight makes the Fourier coefficients of the test function decay like k^{-2} . Over $\mathbb{F}_q[u]$ the analogous statement, that the off-diagonal tends to zero as $q \rightarrow \infty$ at fixed degree, is a finite family of point counts to which the Lang–Weil estimate applies. Numerically the off-diagonal of $t^2 + 1$ is bounded to 10^7 over $\mathbb{Z}$ and decreases with q over $\mathbb{F}_q[u]$.

1 The target and the reduction

In plain terms. The primes along a polynomial are correlated through the small primes; part I turned the strength of that correlation into an exact identity, whose main part (equal roots) is understood, and whose remaining part (distinct roots modulo d) is a question about how the square roots of D modulo composite d are distributed. That question has a sixty-year-old template, Hooley’s divisor problem for $n^2 + a$; this paper adapts it. The weaker, Cesàro form of the statement is what the argument gives, and it is enough for the leading term of the conjecture.

Notation is that of parts I and II: $f = at^2 + bt + c$ irreducible with discriminant D , $\omega_f(p)$, $W_f(d) = db(d) \prod_{p \nmid d} P_p$ on squarefree d with $|W_f(d)| \asymp \lambda(d)/d$, λ multiplicative with $\lambda(p) = p/(p-4)$ at split primes and $\lambda(p) = 1$ at inert ones, $C = C(f)$, $\Sigma_f(H) = \sum_{h \leq H} (S_f(h) - C^2)$. Part I, Theorem 6: with $\phi(\theta) = \theta - \theta^2$ and $m = m_{ss'} \in [1, d-1]$ the representative of $s' - s$,

$$\sum_{h \leq H} \left(1 - \frac{h}{H}\right) (S_f(h) - C^2) = -\frac{1}{2}C \log H + O_f(1) + \frac{C^2}{H} \text{Off}_f^*(H), \quad \text{Off}_f^*(H) = \sum_{d \geq 2} W_f(d) \sum_{\{s, s'\}} \left[-\frac{m(d-m)}{d} + \frac{d}{2} \left(\phi\left(\frac{s}{d}\right) - \phi\left(\frac{s'}{d}\right) \right) \right] \quad (1)$$

the inner sum over unordered pairs of distinct roots modulo d , and trivially $\text{Off}_f^*(H) \ll H \log H$.

Theorem 1 (target). [TO PROVE] For every irreducible quadratic f there is $\eta > 0$ with $\text{Off}_f^*(H) \ll H(\log H)^{1-\eta}$. Consequently

$$\sum_{h \leq H} \left(1 - \frac{h}{H}\right) (S_f(h) - C(f)^2) = -\frac{1}{2}C(f) \log H + O((\log H)^{1-\eta}),$$

i.e. Conjecture 1 of part I holds in Cesàro form for every quadratic, with the first power of $C(f)$.

1.1 Fourier expansion

Since $\phi(\{x\}) = \frac{1}{6} - \pi^{-2} \sum_{k \geq 1} k^{-2} \cos(2\pi kx)$ (absolutely convergent),

$$\text{Off}_f^*(H) = \Omega_0 - \frac{1}{\pi^2} \sum_{k \geq 1} \frac{1}{k^2} T_k(H), \quad T_k(H) = \sum_d d W_f(d) \cos\left(\frac{2\pi kH}{d}\right) \sum_{\{s,s'\}} \cos\left(\frac{2\pi km}{d}\right), \quad (2)$$

where $\Omega_0 = \sum_d W_f(d) \sum_{\{s,s'\}} \left(\frac{d}{6} - \frac{m(d-m)}{d}\right)$ is independent of H . [check: Ω_0 converges absolutely (it is the second-moment discrepancy of the root differences, $\ll \sum_d \lambda(d) \omega(d)^2 \text{disc}_d$; Kowalski–Soundararajan give $\text{disc}_d \rightarrow 0$ on average, but absolute convergence needs a rate; for quadratics Hooley’s Theorem 1 gives a power saving, which suffices).] The trivial bound $|T_k(H)| \ll H \log H$ comes from $d W_f(d) \asymp \lambda(d)$, so that with $K = (\log H)^A$ the frequencies $k > K$ contribute $\ll H \log H/K$ to (2), and Theorem 1 follows from:

Lemma 2 (A: twisted weighted Weyl sums). [TO PROVE] *There are $\eta, A > 0$ such that for all $X, H \geq 2$ and $1 \leq k \leq (\log X)^A$,*

$$\sum_{d \leq X} \lambda(d) \sum_{\nu^2 \equiv D' \pmod{d}} e\left(\frac{k(\nu + H)}{d}\right) \ll X(\log X)^{1-\eta}, \quad D' = D/a^2,$$

uniformly in H (the phase $e(kH/d)$ is the point), together with the same bound for the moduli in progressions $e \mid d$ with polynomial dependence on e , and the corresponding statement for the large moduli $d > H$ after Hooley’s reflection (Lemma C below).

Why the frequency range is mild. With the sawtooth $\{H/d\}$ of part I’s unsmoothed identity the Fourier coefficients decay like $1/k$ and one would need T_k to decay in k ; the Cesàro weight replaces $1/k$ by $1/k^2$, so it is enough to control $k \leq (\log X)^A$ with any polynomial loss in k . This is the single reason the Cesàro form is within reach and the sharp form is not.

2 What is known about the pieces

Unweighted, untwisted. Hooley [2], Theorem 1: $\sum_{d \leq X} \sum_{\nu^2 \equiv D' \pmod{d}} e(k\nu/d) \ll_k X^\theta \log^2 X$ with $\theta < 1$, by parametrising the pairs (ν, d) through the primitive representations $d = Q(r, s)$ by a finite set of binary quadratic forms of discriminant $4D'$ and reducing to incomplete Kloosterman sums (Weil’s bound). Hooley [3] gives equidistribution without a power saving for any irreducible polynomial; Kowalski–Soundararajan [5] reprove it as a mixing property of the Chinese remainder theorem, with an average discrepancy $(\log X)^{-c}$.

Twisted, unweighted. Hooley’s Lemma 5 is a bound $R^\pm(h, X) = \sum_{k \leq X} \rho(h, k) e(\pm hx/k) \ll \sigma(h) |h| X^\theta \log^3 X$ for the same sums with the phase $e(\pm hx/k)$ built into the parametrisation [transcribe the exact exponent from the original; the extracted text is illegible there]; it is exactly what his Σ_5 , our Off, requires, and it is uniform in x .

Smooth weights, moduli in a progression. Duke, Friedlander and Iwaniec [1], Theorem 1.1: for f smooth on $[Y, 2Y]$ with $|f| \leq 1$, $y^2 |f''| \leq 1$, and either sign of D (their §16), $\sum_{c \equiv 0 \pmod{q}} f(c) \sum_{b^2 \equiv D \pmod{c}} e(hb/c) \ll h^{1/4} (Y + h\sqrt{D})^{3/4} D^{1/8-1/1331}$, via the Kuznetsov formula for half-integral-weight Kloosterman sums. Our phase $e(kH/c)$ has $c^2 |\partial_c^2 e(kH/c)| \asymp (kH/c)^2$, so it satisfies their smoothness condition only for $Y \geq kH$: their theorem covers the moduli $c \geq kH$, not the range $c \leq H$ that carries the trivial bound. For $c \leq H$ one needs Hooley’s method with the phase, or a version of Theorem 1.1 with $y^2 |f''| \leq \Delta^2$ and polynomial loss in Δ [check their proof for the dependence on the derivative bound; the scaling $g(x) = \dots$ in their §11 suggests the loss is polynomial].

Weights. Write $\lambda = 1 * g$ with g multiplicative, $g(p) = 4/(p-4)$ at split primes and 0 otherwise; then $\sum_{d \leq X} \lambda(d) F(d) = \sum_e g(e) \sum_{m \leq X/e} F(em)$, the moduli $e > E$ contribute $\ll E^{-1+\varepsilon} X \log X$, and only $e \leq (\log X)^B$ matter. So Lemma A reduces to the unweighted sums over moduli $\equiv 0 \pmod e$ with polynomial dependence on e (Iwaniec [4] for $D = -1$; Ngo for $D > 0$).

Large moduli. For $d > H$ a pair of distinct roots modulo d has $m \leq H$ only if every prime factor of d divides $hQ(h)$ for some $h \leq H$, $Q(h) = a^2 h^2 - D$; Hooley's reflection writes $d = Q(h)/e$ with $e < Q(h)/H$, turning the large moduli into a sum over the small divisors e of $Q(h)$ (part I, companion note, Prop. A3), which is the same structure with e in place of d and the level-of-distribution input for $\#\{h \leq H : e \mid Q(h)\}$ from [4] and Grimmelt–Merikoski.

3 Plan of the proof of Lemma A

[To be written. Steps: (1) Dirichlet convolution to remove λ ; (2) dyadic decomposition in d ; (3) $d \geq kH$: DFI Theorem 1.1 with the phase absorbed in f ; (4) $d < kH$: Hooley's parametrisation $\nu^2 - D' = d\ell$, $d = Q(r, s)$ primitive, $\nu/d \equiv \bar{s}(\dots)/Q(r, s)$; the phase $e(kH/Q(r, s))$ is slowly varying in (r, s) over the regions where Hooley bounds the incomplete Kloosterman sums, so it can be removed by partial summation at the cost of a factor kH/Y per region unless $Y \geq (kH)^{1/2}$, and below that a genuinely new estimate is needed: this is the open core. (5) Assemble with $K = (\log H)^A$.]

4 Route B: the same statement over $\mathbb{F}_q[u]$

Let $\mathcal{A} = \mathbb{F}_q[u]$, q odd, $f = t^2 - D$ with $D \in \mathcal{A}$ squarefree, and for $N \geq 1$ let $\text{Off}_f(N)$ be the off-diagonal of the identity of part I, Theorem 8 (the sum over the $q^N - 1$ nonzero h of degree $< N$). Part I proved that the moduli of degree $\leq N$ contribute nothing and that the remainder is a finite sum:

$$\text{Off}_f(N) = \sum_{N < \deg d \leq M(N)} W_f(d) A_N(d) - q^N \sum_{\deg d > N} \frac{W_f(d)(\omega_f(d)^2 - \omega_f(d))}{|d|}, \quad M(N) = N - 1 + \max(2N - 2, \deg D)$$

$A_N(d)$ the number of ordered pairs of distinct roots modulo d whose difference has degree $< N$.

Theorem 3 (target). *[TO PROVE] For fixed N and $\deg D$, $\text{Off}_f(N) = O(q^{-1/2})$ as $q \rightarrow \infty$. Hence the function-field form of Conjecture 1, $\sum_h (S_f(h) - C^2) = -C(f)N + O(1)$, holds with $O(1)$ replaced by $O(q^{-1/2})$.*

Reduction to point counts. For each degree $k \in (N, M(N)]$, the sum $\sum_{\deg d=k} W_f(d) A_N(d)$ is a weighted count of triples (d, s, s') with d squarefree monic of degree k , $s^2 \equiv s'^2 \equiv D \pmod d$, $s \neq s'$, and $\deg(\widetilde{s' - s}) < N$. Writing $s' - s = m$ with $d \mid m^2 - 4D$ on the part of d where the roots differ and $d \mid m$ on the rest (part I, Lemma 9), the condition is algebraic in the coefficients of d and m , so the count is $\#V_{k,N}(\mathbb{F}_q)$ for an explicit variety $V_{k,N}$ defined over $\mathbb{F}_q$ (depending on D), weighted by $W_f(d) = P(1)\lambda(d)/|d|$, and $\lambda(d)$ is a product of local factors $1 + O(q^{-\deg P})$ over the prime factors of d . The expected term $q^N \sum_{\deg d > N} W_f(d) \omega_f^\circ(d)/|d|$ is the same count with the probability q^{N-k} in place of the indicator of $\deg m < N$. Theorem 3 therefore follows from the Lang–Weil estimate once one knows, for each of the finitely many (k, N) , that the components of $V_{k,N}$ over $\overline{\mathbb{F}}_q$ have the dimensions and multiplicities that make the main term equal to the expected one; this is a finite algebraic computation for each N , and a uniform statement is the content of Katz-type equidistribution for the Salié sums of the roots over primes of fixed degree.

Remark 4 (a $1/q$ law). The data suggest $\text{Off}_f(N) = -\kappa/q + O(q^{-3/2})$ with κ close to 2 and apparently independent of N (for $N = 1$, $\text{Off}_f(1) = -T_{\text{exp}}(1)$ exactly, since no modulus of degree in $(1, M(1)]$ exists). If true, the main terms of the two point counts in the reduction above agree

$f = t^2 - u$ over $\mathbb{F}_q[u]$: the off-diagonal at fixed N as q grows

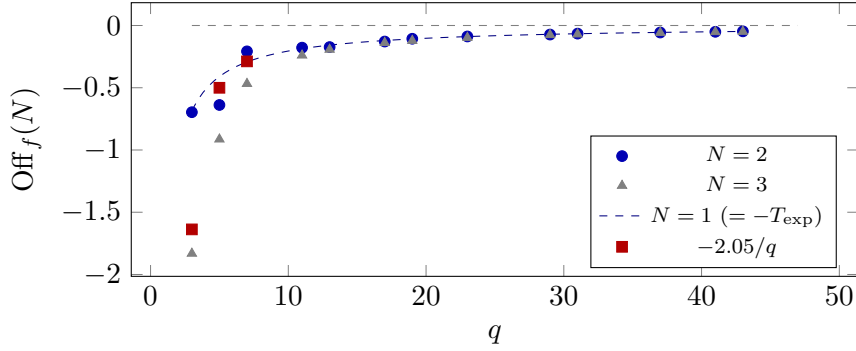


Figure 1: Exact values of $\text{Off}_f(N)$ over $\mathbb{F}_q[u]$ for $f = t^2 - u$ (`scripts/offq.ts`; $N = 2$ for the primes $3 \leq q \leq 43$, $N = 3$ for $q \leq 7$). The decay is faster than the $q^{-1/2}$ of Theorem 3: $q \text{Off}_f(2)$ is $-1.96, -2.25, -2.19, -2.02, -2.03, -2.11, -2.02, -2.08, -2.08, -2.02$ for $q = 11, \dots, 43$, and $q \text{Off}_f(1)$ decreases monotonically towards the same value (-2.13 at $q = 41$).

not only at order $q^{\dim}$ but at order $q^{\dim-1/2}$ as well, and the first discrepancy is at relative order q^{-1} : the varieties $V_{k,N}$ would have point counts polynomial in q to one order beyond Lang–Weil, as rational varieties do. This sharpens the target of Theorem 3 to an asymptotic with an explicit constant, and it is the first place in the programme where the off-diagonal has a *main term* rather than a bound. [compute κ from the degree- $(N+1)$ split primes: the leading contribution to T_{exp} is $q^N \sum_{\deg P=N+1, \omega=2} W_f(P) \cdot 2/|P| \approx 1/((N+1)q)$; the observed $-2/q$ must therefore come from $T_{\text{act}} - T_{\text{exp}}$ at the moduli of degree $N+1$ and $N+2$ jointly, i.e. from the pairs whose difference has degree exactly $N-1$; work it out for $N=1, 2$.]

5 Numerical evidence

Over $\mathbb{Z}$: the off-diagonal of $t^2 + 1$ is bounded to $H = 10^7$, with mean -0.01 , standard deviation 0.51 and no drift in $\log H$ (part II, Section 5.3); a failure of Hypothesis (E) would show as a slope of order 0.73 per unit of $\log H$. Over $\mathbb{F}_q[u]$: Figure 1 and the $1/q$ law of the remark after it.

6 Open issues

- (1) The core of Lemma A: the phase $e(kH/d)$ for $d < (kH)^{1/2}$ within Hooley’s parametrisation.
- (2) The absolute convergence of Ω_0 with a rate.
- (3) Theorem B: identify the components of $V_{k,N}$ for $N=2, 3$ and $\deg D=1$ by hand, then the general pattern.
- (4) Non-2-transitive Galois groups, where a saving over the trivial bound is not enough (part I, Remark 7).

References

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